EXAM 5



116 Question

Physiology:

1- Biochemical change in chronic renal failure include all of the following except :

a. High ALP.
b. High PTH.
c. Low total serum calcitonin
d. Increased troponin
e. Normal ionized calcium

2- fluid replacement therapy in patient who presents with hypovolemia all of the following will occur except :

a. Inhibits ADH secretion
b. Reduced colloid osmotic pressure
c. Increased urine output
d. Low plasma urea
e. Urine Sodium less than 30 mmol/I

3- Hypo secretion of cortisol & aldosterone:

a. Hypopituitarism
b. Pheochromocytoma
c. Addison's disease
d. Con's syndrome
e. Chronic renal failure

4-The following enzyme is useful in diagnosis of osteomalasia:

a. ALP
b. LDH
c. AST
d. Gama glutaml transferase
e. ACP

5- Movement of fluids from the interstitial space to plasma due to:

a. Blood pressure in the capillaries
b. Plasma sodium concentration
c. Plasma potassium concentration
d. Plasma protein concentration
e. Plasma PH

6- Medullary chemoreceptors are more sensitive to :

a. High PCO2
b. High K+
c. Low PO2
d. High PH
e. High H concentration

7-which of the following enzymes not secreted from exocrine pancreas:

a. Lipase
b. Maltase
c. Trypsinogen
d. Ribonuclease
e. Amylase

8- which of the followings inhibits gastric secretion:

a. Static tremor
b. Dysmetria
c. Monotonus speech
d. Gorilla like appearance
e. Lead pipe rigidity

9- Which one of the following would increase in obstructive but not in restrictive long disease:

a. FEV1
b. Breathing frequency
c. Vital capacity
d. Functional residual capacity
e. Maximum breathing capacity

10- Physiologically active thyroxine exists in which of the following forms:

a. Unbound
b. Bound to prealbumin
c. Bound to albumin
d. As a glucuronide
e. Bound to globulin

11- A patient with chronic asthma is started on glucocorticoid therapy. The treatment may result in bone loss because glucocorticoids do which of the following :

a. Suppress vitamin D activation
b. Inhibit bone formation
c. Increase osteoblast growth
d. Inhibit bone resorption
e. Increase calcium absorption form the GI tract

12- Growth hormone excess results in which of the following :

a. Decreased protein synthesis
b. Insulin resistance
c. Decreased lipolysis
d. Decreased gluconeogenesis

e. Hypoglycemia

13-Growth hormone secretion increased by :

a. Hyperglycemia
b. Exercise
c. Hypothermia
d. FFA
e. Somatostatin

14- which one is correct about spermatogenesis:

a. Production and relase of spermatozoa is cyclical
b. Sertoli cell is require for mitotic and meiotic activity for germ cell
c. Spermatogenesis need continuous release of GRH
d. Ledig cell secretion of testosterone require FSH
e. LH act directly on sertoli cell to promote cell devision

15- Thyroid hormone receptor :

a. Has an associated co-repressor protein
b. Does not involve binding of co-activators to initiate gene transcription
c. Is associated with heat shock protein-90 in the cytoplasm
d. Is found free in the cytoplasm and move to the nucleus following its stimulation
e. Is bound to HRE(hormone receptor elements) before hormone binding

16- An abdominal CT in patient with conn's syndrome show multiple small adrenocortical masses which of the following clinical finding is most likely present:

a. Hypertention
b. Hyperkalemia
c. Increased concentrating ability of the kidney
d. Decreased extra cellular fluid volume
e. Increased hematocrit

17-Suckling in lactating lady stimulated which of following:

a. ADH
b. Oxytocin
c. Prolactin
d. ACTH
e. FSH

18- 30 years old lady breast feeding her infant which of the following is expected:

a. Increase secretion of ADH
b. increase secretion of ADH from paraventricular
c. increased secretion of oxytocin from paraventricular
d. decreased secretion of neurophysin

19- which is true about oxytocin hormone:

a. Secreted by supra optic nucleus
b. Induce milk production
c. Stimulated by fear
d. Has trophic effect on breast tissue
e. Concenteration increase at time of delivery

20-Blood pressure equal:

a. COP X Peripheral resistance
b. COP X SV
c. HR X weight
d. HR X SV

21-Parathyroid hormone:

a. Hypersecretion occur in chronic renal failure
b. Is hypocalcaemic hormone
c. Is essential for life as insulin
d. Secretion represent neural hormone control

22-regarding the ca and bone which is true:

a. Filterable fraction is 50% of serum ca
b. Osteoclast has no receptor for parathyroid
c. Osteoporosis can be caused by vit D hypervitaminosis
d. acidosis can cause tetany
e. calcitonin cause hypophosphataemia

23- Cholystyramine :

a. Increase bile acid absorption at distal ileum
b. Enhance bile acid reuptake by liver
c. Act as resin for bile acid
d. Is synthesize at terminal ileum
e. Circulate between liver and terminal ileum

24- C-peptide:

a. Can be isolated from insulin in blood
b. is not detected in type 2 diabetes
c. is found only in pre-proinsulin
d. has antigenic property for insulin
e. enhance insulin uptake by adipocyte

25- Neuromuscular junction is :

a. excitatory
b. excitatory and inhibitory
c. inhibitory
a. b. synapse

26-The first sign of ovarian maturation is :

a. Ovulation
b. Development of breast tissues
c. Appearance of libido
d. Increases body temperature
e. Menstruation

27- FSH is :

a. Lipoprotein hormone
b. Has no role in ovulation
c. Has luteinizing effect
d. Its beta chain is specific
e. Agonized by prolactin

28- Addison's disease characterized by :

a. Hyperglycemia
b. Low ACTH if it is primary
c. Hypokaleamia
d. Normocytic , normochromic anaemia
e. Hyperpigmintaion if it is secondary

29- Which is false about T 4 hormones :

a. Has the same effects , but less strength than T 3
b. Has intracellular receptors
c. Synthetized in follicular cells
d. Does not affect O 2 by the lymphatic tissue
e. Where it acts in the target cells it must deiodonised to T 3

30- The effect of insulin on glucose transport is to :

a. Enhance transport across the tubular epithelium of the kidney
b. Permit transport against a concentration gradient
c. Enhance transport through the intestinal mucosa
d. Enhance transport into the brain
e. Enhance transport into adipocytes

31- Malabsorption assessed by all of the following except :

a- Weight loss
b- Acid base status
c- Anaemia
d- Mild oedema
e- Presence of oesteomalasia

32- woman has tumor secreting large amount of oestrogen. which of the following will occur:

a. Progesterone level in the blood will be very low
b. Her leutenizing hormone secretion rate will be totally suppressed
c. She is not have normal menstrual cycle
d. Her bone will be normally calcified
e. All of the above

33- 22 years old lady presented with amenorrhoea and her investigation show very low level of progesterone which of the following will occur:

a- Low eosterogen
b- High FSH and LH
c- No corpus luteum

34-which hormone is responsible of preovulatory phase:

a- Esterogen

b- Progesteron
c- LH
d- FSH
e- Prolactin

35-sinus bradycardia is:

a- HR more than 100
b- HR less than 60
c- HR more than 60
d- HR less than 100
e- HR more than 50

36- Which of the following is the cause of hypertonicity of the medulla:

a- Diffusion of the water in ascending loop of henle
b- Active reabsorption of sodium from vasarecta
c- Diffusion of water in the ascending loop of henle
d- Reabsorption of water at collecting duct

Anatomy:

37- A 60 years lady was known to have cancer of the ovary presented over the medial aspect of the thigh. The nerve most likely to be :

a. Lumbosacral trunk
b. Medial cutaneous nerve of the thigh
c. Obturator nerve
d. Femoral nerve trunk
e. Intermediate cutaneous nerve of the thigh

38- Venous drainage of lower limb is mainly by :

a. short saphenous vien
b. great saphenous vien
c. deep vien
d. perforators
e. superficial viens

39- During hernial repair internal spermatic fascia is exposed to see ductus deferens and testicular vessels .from which of the following fascia is internal spermatic cord :

a. Internal oblique aponeurosis
b. Processus vaginalis
c. Transversus abdominis aponeurosis
d. External oblique aponeurosis
e. Transversalis fascia

40- Both ischiorectal (ischioanal) fossae communicate :

a. Superiorly
b. Medially
c. Posteriorly
d. Anteriorly
e. Laterally

41- Cancer from pelvic viscera can deposits cancerous cells in the vertebrae this communication occur between the internal iliac veins and the vertebral viens the through the :

a. Lateral sacral veins
b. Lumbar vein
c. Prostatic plexus
d. Umbilical veins
e. Pudenal vein

42- Which of following muscles does NOT attach to the perineal body :

a. Transverse perineal muscles
b. Pubovaginalis
c. External anal sphincter
d. Levator ani
e. Puborectalis

43- In a patient with small volume of fluid collection in the abdomen likely site for fluid collection in the erect position is:

a. Iesser sac
b. Greater sac
c. Douglas pouch
d. Rt hepatorenal pouch
e. Paraduedenal recesses

44- Concerning urinary bladder

a. Against fascia of levator ani
b. Detruser muscle supplied by sympathetic motor fibre
c. Connected to remnant of cloaca
d. In female is completely pelvic organ
e. Its sphincter vesicae has parasympathetic motor supply

45- Muscle of urogenital diaphragm include:

a. Sphincter urethrae
b. Sphincter vesicae
c. Levator ani
d. Ischiocavernosus
e. Obturator internus

46- Regarding the anterior abd wall:

a. Superficial tissues of the anterolateral wall above theumbilicus to axillary nodes
b. Intertubercular plane lies at the level of L4
c. Mid-inguinal point lying halfway between the pubic tubercle and ASIS
d. Posterior wall of the inguinal canal is formed by the transversalis muscle
e. Sugrical incision at linea alba heals faster than that along paramedian abd wall

47- Concerning the peritoneum:

a. Right sub-hepatic pouch also called the lesser sac
b. The medial umbilical ligament contains the urachus
c. It's a mucous membrane
d. Left sub-hepatic pouch is the hepatorenal pouch of morris
e. The epiploic foramen is bounded anteriorly by the portal vein

48- The pelvic part of the ureter :

a. Is related to the ischial spine before it turns medially toward the bladder
b. Is an intraperitoneal structure
c. Enters the false pelvis in front of the bifurcation of the common iliac vessels
d. Enters the bladder by passing directly through its wall
e. Is supplied by the iliac vessels

49- The femoral sheath is formed by which of the following :

a. The fascia lata and the membranous layer of the superficial fascia
b. The pectineus fascia
c. The fascia iliaca and the fascia transversalis
d. The psoas fascia and the fatty layer of superficial fascia
e. The processus vaginalis

50- The oviducts are derived from the :

a. Mesonephric ducts
b. Cloaca / urogenital
c. Peritoneum
d. Metanephricblastema
e. Paramesonephric ducts

51- pudendal nerve supply all the fallowing except :

a. Skin of the penis
b. Sphincter Urethrae
c. Ischiocavernus muscle
d. Posterior vaginal fornix

Biochemistry:

52- A young child was diagnosed as a case of failure to thrive (failure to grow) because of essential amino acid deficiency. Which of the following amino acids cannot in treatment even when supplied in adequate amounts :

a. Valine
b. Isoleucine
c. Leucine
d. Methionine
e. Tyrosine

53- Histidine catabolism :

a. Is related to folate deficiency anemia
b. Can lead to increased foraminoglutamic acid secretion in urine B12 deficiency
c. Does not involves THF
d. Produces aspartate as an end product
e. Uroconate is not produced

54- Alkaptonuria :

a. Is associated with lens dislocation
b. Homogentisate accumulation in the kidney leads to renal failure
c. Is an X-linked recessive disorder

d. Benzoquinone acetate accumulation causes ochronosis
e. Is a life threatening disease

55- Phenylketonuria Type-II and III :

a. Is more common in males
b. Due to lack of NADPH
c. Results from dihydrobiopterin deficiency
d. Dihydrobiopterin reductase activity is low
e. Associated with a negative ferric chloride urine testing

56- Beriberi :

a. Is not common in alcoholics
b. The deficient vitamin never affect the brain
c. Is the deficiency of riboflavin
d. Affects the nerves and the heart
e. Is the combination of diarrhea, dementia and dermatitis

57- Pantothenic acid :

a. Form part of S-Adenosylmethionine
b. Needs a carrier in plasma
c. Coenzyme in transamination
d. Can cause burning foot syndrome
e. Is not commonly found in food

58- Hormone receptors :

a. Are proteins
b. Are cytosolic
c. Effective binding and action occurs within the physiological level
d. Are nuclear
e. Are membrane bound

59- Isoenzymes :

a. Are functional plasma enzymes
b. Are 5 for creatine kinase
c. Differ in affinity to substrate
d. Are not used in diagnosis
e. Have different functions

60- The key reaction in glycolysis is catalyzed by :

a. Phosphofructokinse-1
b. Glucokinase
c. Pyruvate kinase
d. Fructokinase
e. Phosphohexoseisomerase

61- The citric acid cycle :

a. Does not participate in amino acid synthesis
b. Is the primary site for CO2 production
c. Dose not produce ATP at substrate level
d. A cytosolic pathway
e. Is amphipathic

62- Gluconeogenesis :

a. Is a mitochondrial pathway
b. Occurs in the liver only
c. Produces glucose for insulin dependent tissues
d. Involves the utilization of amino acids and glycerol
e. Does not occur in the liver in the fed state

63- Glycogen phosphorylase :

a. Activity increases in the presence of high amount of insulin
b. Is activated by dephosphorylation
c. Produces uridine diphosphate from glycogen
d. Produces glucose -1- phosphate
e. Is a mitochondrial enzyme

Genetics:

64- Synthetic nucleotide analogs :

a. Can inhibit DNA synthesis
b. Are always modified purine nucleotides
c. Lack phosphate group
d. Allow replication to proceed at a faster rate
e. They cause denaturation of DNA

65- Pseudouridine is :

a. Not part of the D arm of trna
b. A synthetic uridine nucleotide
c. Based paired to adenine nucleotide in mrna
d. A modified uridine in which methyl group is added to trna

e. A temporary form of uridine that changes during transcription

66- The coding strand of DNA is:

a. Part of the template strand
b. Contain
c. uracil instead of thymine
d. The strand that is complementary to the template strand
e. Contain uracil instead of thymine
f. Identical in sequence to mrna that codes for proteins

67- Nucleosome:

a. Contain negatively charged protein
b. Is coiling of each single strand of DNA and histone
c. Is association of each DNA and arginine rich protein
d. Is combination of DNA and chromatin
e. Is association of DNA ,RNA,and non histone protein

68- mRNA cap:

a. Is not required for translation
b. Is not found in all mrnas that code for synthesis of different proteins
c. Consist of 25 to 250 nucleotides
d. Protect against 5 exonuclease
e. Is 7 formyl GTP

69- Major and minor groove in DNA :

a. major and minor grooves are site for polymerase enzymes.

Pathology:

70- One of the following is example of hypertrophy due to hormonal change:

a- Uterine fibroid
b- Uterus change due to menstrual cycle
c- Acromegaly
d- Muscle of atheletes
e- Uncontrolled hypertention

71- 68 years old man c/o recent change in bowel habit s show bloody stool colonoscopy reveal adenocarcinoma. the malignancy acquired set of mutations that cause which of the following behavior:

a- Increased cell-cell adhesion
b- Increased susceptibility to apoptosis
c- Loss of cell cycle point control
d- Enhance stem cell differentiation
e- Decreased cellular motility

72- Which one of the following for medullary carcinoma of thyroid gland :

a- T 4
b- TSH
c- Calcitonin
d- Low urinary iodine
e- Thyroglobulin

73- Which of the following indicates defect in synthetic function of the liver:

a- Hyperbilirubinaemia
b- High AST
c- Prolonged prothrombin time
d- Vit k deficieny
e- High ALP

74- On x-ray done for 65 years old patient from suffering schistosomiasis small areas of calcification in his liver this calcification is:

a- Is sign for hypercalcaemia
b- Indication of liver cirrhosis
c- Indication of liver fibrosis
d- Is normal finding of such age
e- Indication of presence of Schistosoma ova

75- 23 years old man c/o calves pain after walking is relieved by rest which underlying cause is

a- Peripheral neuropathy
b- Acute osteomyelitis
c- Polymyositis
d- DVT
e- Atherosclerosis

76- Generalized oedema is seen in :

a- Congestive heart failure
b- DVT
c- Filiarsis

d- Cushing syndrome
e- SIADA

77- Metastatic calcification occur in:

a- Acute pancreatitis
b- Primary hyperparathyroidism
c- Secondary hyperparathyroidism
d- TB
e- Duodenal ulcer

78- Tumor cell is :

a. Self but not altered
b. Self cell
c. Non self cell
d. Self altered cell
e. A target cell for T helper cell

79- The normal immunological control of tumors is referred to :

a. Super antigen recognition
b. Immunological tolerance
c. Immunological surveillance

Microbiology:

80- Infected by HIV virus is :

a. T –helper
b. Sperm
c. T-suppressor
d. T-cytotoxic
e. Monocyte

81- Regarding HIV infection , slow progressors :

a. Have mutation in both genes coding for the coreceptors
b. Are genetically immune against HIV infection
c. Lack CD4 molecules
d. Have mutation in one of the genes coding for CCR, and CXCR
e. Have efficient immune system that eliminates the virus

82- The following is NOT correct about influenza viruses :

a. Type A undergoes antigenic shift and drift
b. Grows in the amniotic and as well the allantoic cavity of egg
c. produce 9 types of neuraminidases
d. Does not cause CPE on tissue culture
e. Type B undergoes antigenic shift only

83- The following virus has one DNA strand :

a. Parvovirus
b. Rotavirus
c. Papiloma virus
d. Hepatitis C virus
e. HIV virus

84- The following are correct about Rhinovirus EXCEPT :

a. Causes diffuse rounding of cells
b. Icosahedral
c. Non-enveloped
d. Prevented by a live attenuated vaccine
e. Detected in tissue culture by hemagglutination test

85-The following virus is a double stranded RNA virus …. :

a. HIV virus
b. Hepatitis E virus
c. Hepatitis B virus
d. Rota virus
e. Herpes simplex virus

86- Fungi pose a problem in treatment because they are :

a. Can assimilate any substance
b. Eukaryote
c. Prokaryotes
d. Partially anaerobic
e. Heterotroph

87-Which of the following is the commonest site of paranasal aspergillosis:

a. Orbit
b. Buccal
c. Maxillary sinus
d. Frontal sin
e. Ethmoidal sinus

88- Systemic fungal infections are common in :

a. Diabetics
b. Critically ill
c. Unvaccinated
d. Immunocompromised patients
e. Malnourished patients

89- Which of the following is completely cured by surgical removal:

a. Paranasal aspergilloma
b. Allergic aspergillosis
c. Otomycosis
d. Invasive aspergillosis
e. Pulmonary aspergilloma

90-A woman aged 40 years c/o non offensive discharge like milk the most propaple organism is:

a- Gardenella vagialis
b- Cryptococcus neofomans
c- Candida albicans
d- Trichomonas vaginalis

91-Which of the following tests will give you a correct answer the same day:

a- Culture on sabourauds agar
b- Fluorescent antibody test
c- Histopathology
d- Gram stain
e- Prepare specimen in 20%KOH

92-The commonest cause of fungal sinusitis in sudan is:

a- Flavus
b- A.nidulans
c- Aspergillus fumigatus
d- Rhizpas oryzae
e- Absidia ramose

93-Which of the following antibiotics is the least effective on pseudomonas aurogonsa:

a- Ticarcillin
b- Azlocillin
c- Piperacillin
d- Amoxicillin
e- Gentamycin

94-Which of the following will unequivocally confirm urinary tract infection :

a- Burning micturition
b- Dysuria
c- >10 pus cells /high field
d- Clinical symptoms
e- Significant bacteuria

95-The single most important characteristic of diarrhea caused by vibrio cholera:

a- Severe abdominal pain
b- Massive bloody diarrhea
c- Profound watery diarrhea
d- Renal insufficiency

96-Infections caused by bacteroid fragilis are best treated by:

a- Cefoxitim
b- Chloloramphenicol
c- Clarithromycin
d- Metronidazole
e- Gentamycin

97-which of the following organim is not cause of nosocomial infection:

a- pseudomanas aerogonosa
b- kleibsella
c- Enterobacter
d- E.coli
e- Streptococcus pyogen

98-An important means of preventing hospital acquired infection:

a- For all patient to be placed
b- For all patient to receive asingle dose of antibiotics at time of presentation
c- For health care patients to wash their hands between patients
d- An infection control program with one nurse for every 10 patients

99-The following are cause of gonococcal urethritis except:

a- Mycoplasma genitalium
b- Mycoplasma hominis
c- Chlamydia pneumonia
d- Ureaplasma urealyticum
e- Chlamydia trachomatous

100-An icosahedral virus is the virus which :

a. Lacks the capsid layer
b. Resist to ether
c. Duple layer capsid
d. RNA
e. Their capsid arranged in cubes

Immunity:

101- The classical pathway for complement activation is :

a. C5 –C9
b. An antigen – antibody reaction
c. Factors released from damaged tissues
d. Factors released from phagocytes
e. A foreign polysaccharide

102- Immunity obtained from the passage of antibodies across the placenta is :

a. Species immunity
b. Natural passive immunity
c. Artificial passive immunity
d. Natural active immunity
e. Artificial active immunity

103- Passive immunization :

a. Is not effective
b. Acts immediately
c. Needs T and B cells interaction
d. Revert to virulent
e. Gives permanent protection

104- The following contain attenuated live vaccines EXCEPT :

a. BCG
b. TAP
c. Measles
d. Yellow fever
e. Polio (sabin)

105- Which of the following abnormalities is best described as a secondary immunodeficiency :

a. Lack of helper T lymphocytes
b. Failure to make the IgM, IgE and IgG
c. Failure of T cell to mature 2ry to structural and functional alteration of the gland
d. Selective deficiency of IgA production
e. Immunodeficiency following radiation exposure

Pharmacology:

106- A 45-year – old female treated for ovarian cancer develops hearing loss of the following agents most likely caused these finding :

a. Doxorubicin
b. 5-FU
c. Cisplatin
d. Bleomycin
e. Paclitaxel

107- A positive coombs' test and hemolytic anemia may following which antihypertensive drug :

a. Prazosin
b. Methyldopa
c. Clonidine
d. Guanabenz

108- The most effective agent in the treatment of rickettsia mycoplasma and chlamydia infection is :

a. Gentamicin
b. Penicillin G
c. Bacitracin
d. Vancomycin
e. Tetracycline

109- A nucleophilic attack on deoxyribonucleic acid which cause disrubtion of base pairing occurs as a result of :

a. 5-FU
b. Thioguanine
c. Prednisone
d. Methotrexate
e. Cyclophosphamide

110- Cardiotoxicity limits the clinical usefulness of which one of the following antitumor antibiotics :

a. Doxorubicin
b. Bleomycin
c. Dactinomycin
d. Plicamycin
e. Mitomycin

111- Inhibitors of serotonin (5-HT) such as paraxetin interact with which of the following:

a. Chlorpromazine
b. Halothane
c. Benztropine
d. Digoxin
e. Tranylcypromine

112- The skeletal muscle relaxant that act directly on contractile tissues of muscle fibre is:

a- Baclofen
b- Pancuronium
c- Gallamine
d- Dantrolene
e- Cyclobenzaprine

113- Clavulanic acid is important because it :

a. Is a potent inhibitor of cell-wall transpepitdase
b. Has a spectrum of activity similar to that of penicillin
c. Is specific for gram-positive microorganisms
d. Easily penetrates gram-negative microorganisms
e. Inactivates bacterial B-lactamases

Statistics:

114- Gradual decrease in x-ray beam intensity as it passes through is called :

a. Decay
b. Impedence
c. Imaging
d. Attenuation
e. Radioactivity

115- Speed of ultrasound depends upon :

a. Material
b. Size of the transducer
c. Medium
d. Wavelength
e. Amplitude

116- Which of the following is CORRECT :

a. A large sample size always ensures that our sample size is accurate and precise
b. In random samples the randomization ensure that stimate
c. We do not need to randomize if our sample size is sufficiently enough
d. If all other things are equal , we need a larger sample size

In a properly chosen sample , an estimate will be less and more precise